

SREYA C V

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SUMMARY

B.Tech Bioinformatics student (Class of 2027) who builds data and machine-learning pipelines end to end: acquiring public datasets, cleaning and representing them, training and validating models, and packaging the result so it runs with one command. Built a retrieval-augmented generation service with a provider-agnostic inference layer, and a reproducible multi-omics analysis pipeline now prepared for journal submission. Looking for a DevOps / MLOps role where reliable data preparation, deployment and model monitoring matter.

TECHNICAL SKILLS

Languages: Python (primary), R, C, JavaScript/TypeScript, SQL, shell

ML / NLP: scikit-learn, sentence-transformers, LangChain, vector search, embeddings, RAG, evaluation (CV, C-index, decision curves), PyTorch basics

Data & pipelines: reproducible one-command pipelines, public-data ingestion (TCGA, GEO, GTEx, ChEMBL), batch-effect diagnosis, Git, Linux

Serving: Streamlit, OpenAI-compatible LLM endpoints (Ollama, vLLM), streaming responses

Fundamentals: OOP, data structures & algorithms, biostatistics

PROJECTS

RAG-Based Document Q&A Assistant

github.com/sreyacv1/q-n-a-assistant

Python | LangChain | sentence-transformers (all-MiniLM-L6-v2) | vector store | Ollama / vLLM | Streamlit

- Built the full pipeline: PDF/text ingestion, chunking, embedding, semantic retrieval, and streamed grounded answers with inline [n] citations and relevance-scored sources.
- Designed a provider-agnostic LLM layer so the same code serves from a local Ollama model or a hosted vLLM endpoint, switching by configuration only.

Gastric Cancer Multi-Omics Pipeline — Prognosis, CAF Programme & Tumour Microbiome

github.com/sreyacv1/GASTRIC-CANCER

R 4.3 | TCGA-STAD, GTEx, 5 GEO cohorts, scRNA-seq, 16S microbiome, IEU OpenGWAS | LASSO-Cox, WGCNA, GSEA, Mendelian randomisation

- Acquired, harmonised and versioned eight public datasets into a reproducible one-command R pipeline; trained a 25-gene LASSO-Cox prognostic model with nested cross-validation (C-index 0.61) and validated it on independent external cohorts.
- Diagnosed an apparent microbiome “dysbiosis” signal as a sequencing-batch artefact through drift analysis across cohorts; reported an honest null from two-sample Mendelian randomisation. Manuscript prepared for journal submission.

Cracking KRAS G12D — AI-Guided De Novo Inhibitor Design

Boltzmann AI Hackathon 2026, individual entry

Python | DiffDock | RDKit | ChEMBL / SureChEMBL | PDB 7RPZ

- Generated and filtered 46 de novo molecules against the KRAS G12D switch II pocket with an approved inhibitor (MRTX1133) as positive control through every stage; 8 of 15 candidates beat the reference’s in-pocket pose confidence, reproduced across two independent runs.
- Automated the screen as a scripted pipeline: docking, drug-likeness (Lipinski / QED), and novelty checks against ChEMBL (Tanimoto 0.43–0.67).

Unalone — AI-Powered Emotional Wellness Companion

github.com/sreyacv1/Unalone

Smart India Hackathon, cleared internal college selection | JavaScript | Node.js | LLM chatbot with voice

- Built the frontend: mood-adaptive chatbot interface, anonymous peer connect and one-click personalised actions.

DuoGrow — AI Accountability OS for Two

github.com/SaiK1105/duogrow

OpenAI Hackathon, “AI in Daily Life” track | TypeScript

- Team lead. Owned the concept, split the work across the team and drove the build to a working hackathon submission.

EDUCATION

B.Tech, Bioinformatics — SASTRA Deemed to be University, Thanjavur

Expected 2027

Data Structures & Algorithms, Machine Learning, Deep Learning, Big Data, Biostatistics, Genomics & Proteomics

Class XII & X — MES Central School, Tirur, Kerala

78.4% (2021) | 89.4% (2019)

Languages: English (fluent) | Malayalam (native) | Tamil | Hindi